

Astrophysics III — Answers to All Listed Practice Questions

Organized by chapter. Each entry: **Q** → **A** (precise, exam-ready).

CHAPTER 4 — Exoplanet Detection

4.1 Orbits

Q1. Derive Kepler's second and third laws from the two-body equation of motion.

From  $\ddot{\mathbf{r}} + G(M_1 + M_2)\mathbf{r}/r^3 = 0$ , the force is central ( $\mathbf{r} \times \mathbf{F} = 0$ ), so  $\mathbf{r} \times \dot{\mathbf{r}} = \mathbf{h} = \text{const.}$  In polar coordinates this gives  $r^2\dot{\Theta} = h$ . Since the swept area is  $\delta A = \frac{1}{2}r^2\delta\Theta$ , we get  $\dot{A} = h/2 = \text{const}$  — **Kepler's 2nd law** (equal areas in equal times).

For the 3rd law: solving the radial equation via Binet's substitution  $u = 1/r$  gives the ellipse  $r = a(1 - e^2)/(1 + e \cos f)$ , with  $h^2/[G(M_1 + M_2)] = a(1 - e^2)$ . Using  $A = \pi ab$ ,  $b^2 = a^2(1 - e^2)$ , and  $P = 2A/h$ :  $h = \frac{2\pi}{P}a^2\sqrt{1 - e^2} \Rightarrow \boxed{P^2 = \frac{4\pi^2}{G(M_1 + M_2)}a^3}$

Q2. What are  $e, \omega, i, \Omega$  physically?

- $e$  — eccentricity, shape of the ellipse (0 = circle).
- $\omega$  — argument of periapsis: angle (in the orbital plane) from the ascending node to periapsis.
- $i$  — inclination: angle between the orbital plane and the reference plane (sky plane).
- $\Omega$  — longitude of ascending node: angle (in the reference plane) from a fixed reference direction to the line of nodes.

Q3. Why does  $\Omega$  not appear in  $Z(t)$  but does in  $X, Y$ ?

$Z(t) = r(t) \sin(\omega + f) \sin i$  (Eq. 4.23).  $\Omega$  only rotates the orbit *within* the sky plane ( $X, Y$ ) about the observer's line of sight ( $Z$ -axis); it does not tilt the orbit toward/away from the observer. Since  $Z$  is the line-of-sight coordinate, it is invariant under rotations about that same axis — hence  $\Omega$  drops out of  $Z$ .

Q4. Why is Kepler's equation solved iteratively?

$M(t) = E(t) - e \sin E(t)$  is **transcendental** in  $E$  (mixes  $E$  and  $\sin E$ ) — there is no closed-form inverse in elementary functions. It is solved numerically, e.g. Newton's method starting from  $E_0 = M_0$ .

Q5. Which orbital elements can RV alone give you?

RV gives  $P, e, \omega, t_0$  (periapsis time), and  $K_1 \Rightarrow M_2 \sin i$ . It **cannot** give  $\Omega$  (no sensitivity to rotation about the line of sight) nor  $i$  separately from  $M_2$  (only the combination  $M_2 \sin i$ ) — true mass and inclination are individually degenerate from RV alone.

4.2 Radial Velocity

Q6. Derive  $K_1$ ; why does it scale via Kepler's 3rd law?

Differentiating  $Z(t) = r \sin(\omega + f) \sin i$  w.r.t. time (using  $\dot{r}$  from Eq. 4.15 and  $h = r^2\dot{f}$ ) gives:  $v_{z1} = v_r + K_1[\cos(\omega + f) + e \cos \omega]$ ,  $K_1 = \frac{2\pi a \sin i}{P \sqrt{1 - e^2}} \frac{M_2}{M_1 + M_2}$ . Using Kepler's 3rd law,  $a \propto [(M_1 + M_2)P^2]^{1/3}$ , so for  $M_2 \ll M_1$ :  $K_1 \propto M_2 \sin i P^{-1/3} M_1^{-2/3}$  — heavier/closer-in planets around lighter stars give the largest signal.

Q7. Why can we never measure an absolute RV? Role of  $\Phi/c^2$ ?

The observed wavelength includes gravitational redshift:  $\lambda = \lambda_0 \frac{1 + \frac{1}{c}\mathbf{k} \cdot \mathbf{v}_*}{1 - \Phi/c^2 - v_*^2/2c^2}$ .  $\Phi$  (gravitational potential at the stellar surface) is essentially constant ( $\sim 0.5$  km/s for a dwarf) and unknown/unmeasurable precisely — it acts as an unremovable *offset*. Only the **shape/amplitude** of the RV curve (relative variation,  $K_1$ ) is meaningful, not its absolute zero-point.

Q8. Why do narrow/strong lines give better RV precision (via  $Q$ )?

Precision is  $\sigma_{RV} = c (\sum_i A_i)^{-1/2} Q^{-1}$ , with  $Q^2 = (\sum_i \lambda_i^2 (\partial A_i^0 / \partial \lambda_i)^2 / A_i^0) / \sum_i A_i$ .  $Q$  weights pixels by the **steepness**  $(\partial A / \partial \lambda)^2$  of the line profile — narrow, deep lines have steep slopes on their flanks, so a tiny Doppler shift produces a large, well-measurable flux change there. Shallow/broad lines shift the same amount but change flux far less per pixel → worse precision.

Q9. Jupiter vs Earth RV amplitude; why hot Jupiters first?

Jupiter at 5 AU:  $K \approx 12.7$  m/s. Earth at 1 AU:  $K \approx 9$  cm/s — about  $140\times$  smaller. Early instruments only reached  $\sim 10$  m/s precision, so only giant, close-in planets (hot Jupiters,  $K \sim 100$ s of m/s, short  $P$ ) were detectable — both because  $K \propto M_2/P^{1/3}$  favors them, and because short periods mean full orbits (and thus confident detections) could be observed in a few years.

Q10. Physical line broadening vs.  $v \sin i$  broadening?

Physical (intrinsic) broadening — natural, Stark, Van der Waals, thermal Doppler motion — sums as Gaussian widths,  $\gamma = \sum_i \gamma_i$ , and is symmetric/local to each atom.  $v \sin i$  broadening is *not* intrinsic to the line: it's the superposition of many Doppler-shifted copies of the same line profile from different points across the rotating stellar disk (a convolution  $H(\lambda) \star G(\Delta\lambda)$ ), producing a broad, often flat-bottomed profile whose width directly measures projected rotation speed.

Q11. Why is  $M_2 \sin i$  a minimum mass? How far off from  $M_2$ ?

Since  $\sin i \leq 1$ ,  $M_2 \sin i \leq M_2$  always — hence "minimum mass." For isotropically oriented orbits,  $P(i) di = \sin i di$ , with  $\langle \sin i \rangle = \pi/4 \approx 0.79$ , and the distribution is skewed toward  $i$  near  $90^\circ$  (edge-on) — most systems are seen close to edge-on, so statistically  $M_2 \sin i$  is usually within a factor of  $\sim 2\text{--}3$  of  $M_2$ , and often a reasonable direct proxy.

Q12. Brown dwarf vs. giant planet confusion (HD 114762); why rare?

Without knowing  $i$ , a large  $M_2 \sin i$  (e.g. 11  $M_J$  for HD 114762) could mean a genuine massive planet at high  $i$ , **or** a much more massive object (stellar/brown dwarf) seen nearly pole-on (small  $\sin i$ ). Astrometry later confirmed HD 114762's companion is stellar. This confusion is statistically rare because brown dwarfs are intrinsically uncommon companions ("brown dwarf desert"), so the joint probability of (a) picking a rare object *and* (b) it being viewed at a rare near-pole-on geometry is small — most  $M_2 \sin i \sim \text{few } M_J$  detections really are planets.

4.3 Transits

Q13. Derive the transit probability; why easier for hot Jupiters?

A transit occurs if  $\sin \Theta = (R_* \pm R_p)/r_{\text{tra}}$  where  $\Theta$  is the max misalignment allowing an eclipse. Averaging over random  $\omega$ :  $\langle P \rangle = \frac{R_* \pm R_p}{a} \frac{1}{1-e^2} \approx \frac{R_*}{a}$ . Since  $1 \text{ AU} \approx 200 R_*$ ,  $P \approx 0.5\%$  at 1 AU but  $\approx 10\%$  at 0.05 AU (51 Peg b-like) — close-in planets subtend a much larger solid angle as seen from the star, so alignment is far more likely.

Q14. Why does transit duration depend only on stellar density & period?

$\Delta T = T_0 \sqrt{1-b^2}$ ,  $T_0 = \frac{R_p P}{\pi a}$  Using Kepler's 3rd law  $a \propto M_*^{1/3} P^{2/3}$  and  $M_* \propto \rho_* R_*^3$ :  $T_0 \propto \rho_*^{-1/3} P^{1/3}$  — the planet's radius/mass never entered the derivation (it came purely from stellar geometry + orbital velocity), so  $T_0$  measures **stellar density**, independent of the planet.

Q15. Derive  $\Delta T = T_0 \sqrt{1-b^2}$ .

For a circular orbit with  $\omega = \pi/2$  (mid-transit at  $f = 0$ ):  $x(t) = a \sin f$ . The in-transit chord length is  $\Delta x = 2R_* \sqrt{1-b^2}$  (chord of the stellar disk at impact parameter  $b$ ), giving  $\Delta f = \sin^{-1}(\frac{2R_*}{a} \sqrt{1-b^2})$ . With  $\Delta T = \frac{P}{2\pi} \Delta f$  and  $R_*/a \ll 1$  (small-angle):  $\Delta T = \frac{R_* P}{\pi a} \sqrt{1-b^2} = T_0 \sqrt{1-b^2}$ .

Q16. How to get  $b$  from a light curve?

Combine total transit duration  $\Delta T = T_0 \sqrt{1-b^2}$  with ingress/egress duration  $\tau$  (time to fully occult the star, i.e.  $2R_* \sqrt{1-b^2}/\Delta T = 2R_p/\tau$ ). Rearranged:  $\delta \cdot (\Delta T/\tau)^2 = 1-b^2$ , giving  $\tau = T_0 \sqrt{\delta}$ . So measuring  $\Delta T$ ,  $\tau$ , and depth  $\delta$  jointly solves for  $b$ .

Q17. What causes limb darkening; why wavelength-dependent?

Limb darkening arises because we see progressively hotter, deeper photospheric layers toward the disk center vs. cooler, shallower layers near the limb (where the line of sight is more oblique). Since the emergent intensity is closer to Planck function  $B_\lambda(T)$ , the contrast between center and limb depends on how strongly  $B_\lambda$  varies with  $T$  at a given  $\lambda$  — this varies with wavelength (stronger in blue/optical, weaker in IR), so limb darkening is wavelength dependent.

Q18. What does transit-occultation timing give you?

$\Delta t_c \approx \frac{P}{2} (1 + \frac{4}{\pi} e \cos \omega)$  Measuring the time between transit and occultation (relative to  $P/2$ ) gives  $e \cos \omega$  — a combination, not  $e$  and  $\omega$  separately (the duration ratio  $\Delta T_{\text{tra}}/\Delta T_{\text{occ}} \approx 1 + e \sin \omega$  would be needed for the other combination, but is observationally harder).

Q19. Why do puffy hot Jupiters show larger transmission-spectroscopy signals?

$\Delta \delta = \frac{(R_p+nH)^2 - R_p^2}{R_*^2}$ ,  $H = \frac{k_B T}{\mu m g}$   $H$  (scale height) is large for **hot** (large  $T$ ) and **low-gravity/low-density** (small  $g$ ) atmospheres — exactly hot Jupiters with inflated envelopes. A larger  $H$  means the wavelength-dependent apparent radius varies more, giving larger  $\Delta \delta$  (up to  $\sim 10^{-4}$ , i.e.  $\sim 1\%$  radius variation).

Q20. Thermal vs. reflected occultation depth — which dominates where?

Thermal:  $\delta_{\text{oc}} = k^2 B_\lambda(T_p)/B_\lambda(T_*) \approx k^2 T_p/T_*$  (Rayleigh-Jeans) — dominates in the **IR**, where the planet's own blackbody emission is relatively bright. Reflected:  $\delta_{\text{oc}} = A_\lambda (R_p/2a)^2$  — dominates in the **visible/optical**, where stellar flux is bright but the planet emits little thermally; requires a large, close-in, reflective planet to be detectable.

Q21. What is the Rossiter–McLaughlin effect; why a distortion, not a shift?

As the planet transits, it sequentially blocks light from the blue-shifted (approaching) and red-shifted (receding) halves of the rotating stellar disk. This *removes* flux asymmetrically from the line profile rather than shifting the whole star's velocity — it's a **line-shape distortion**. A standard RV pipeline (which fits a single Doppler shift to the whole line) still responds to this distortion as an *apparent* time-varying RV anomaly,  $\Delta v_{RM} \approx 0.7 k^2 \sqrt{1-b^2} v_* \sin i_*$ , even though no bulk stellar motion occurred.

Q22. Why do hot stars show misaligned obliquities, cool stars aligned?

Tidal dissipation, which realigns the stellar spin with the orbital plane over time, is efficient in **cool stars** (they have convective outer envelopes, providing strong tidal friction) but weak in **hot stars** (radiative envelopes, inefficient tidal coupling). So cool-star systems have had time to realign ( $\lambda$  small), while hot-star systems retain their (often dynamically excited/misaligned) primordial or post-migration orientation.

Q23. What do transit + RV together give that neither gives alone?

Transit gives  $R_p/R_*$  (and, with stellar models,  $R_p$ ) and — crucially — constrains  $i$  (near  $90^\circ$  for a transit to occur), breaking the  $RV \sin i$  degeneracy. Combined: **true mass**  $M_p$  (not just  $M_p \sin i$ ), and with  $R_p$  known, the planet's **bulk density**  $\rho_p = M_p/(\frac{4}{3}\pi R_p^3)$  — telling you rocky vs. gaseous composition.

CHAPTER 2 — Protoplanetary Disks

Q24. Why can't we see H<sub>2</sub> directly; what's used instead; limitations?

H<sub>2</sub> is a symmetric homonuclear molecule — no permanent dipole moment — so it lacks strong rotational transitions; it's only seen via weak UV fluorescence or IR lines buried under dust continuum. **CO** is used as a proxy (strong dipole, bright rotational lines at 115/230/345 GHz for  $J=1-0, 2-1, 3-2$ ). Limitations: requires assuming a fixed CO/H<sub>2</sub> ratio (which varies, e.g. 10<sup>-4</sup> down to 10<sup>-6</sup>), lines can saturate (become optically thick), and CO freezes out below ~17 K (~150 AU) — useless in the cold outer disk.

**Q25. Why is SED flux  $\propto$  disk mass in the optically thin limit? Where does it break?**

$F_\nu = \frac{\cos \vartheta}{D^2} \int_{r_{\text{in}}}^{r_{\text{out}}} B_\nu(T_d)(1 - e^{-\tau_\nu}) 2\pi r \, dr$  For  $\tau_\nu \ll 1$ :  $1 - e^{-\tau_\nu} \approx \tau_\nu = \kappa_\nu \Sigma_d(r) / \cos \vartheta$ , so the integral becomes  $\propto \int \Sigma_d(r) 2\pi r \, dr = M_d$  (total dust mass), giving  $F_\nu \sim \frac{\kappa_\nu}{D^2} B_\nu(T_d) M_d$ . Breaks down where  $\tau_\nu \gtrsim 1$  (optically thick, typically the inner disk, or wherever  $\lambda$  is not  $\gg$  grain size, roughly  $\lambda \gg 100 \mu\text{m}$  needed).

**Q26. Why is crystalline silicate emission (forsterite) diagnostic?**

Forsterite (Mg<sub>2</sub>SiO<sub>4</sub>) requires  $T > 1000$  K to form — such temperatures only occur very close to the star. Detecting crystalline silicate features (vs. amorphous) therefore signals material that was thermally processed near the star, later perhaps transported outward — a tracer of disk thermal history/radial mixing.

**Q27. Sketch SED evolution primordial  $\rightarrow$  transitional  $\rightarrow$  debris.**

- **Primordial** (classical T Tauri): optically thick, gas-rich, strong CO + sub-mm + near-IR excess.
- **Transitional** (weak-line T Tauri): inner disk cleared (no near-IR excess), outer disk still optically thick (sub-mm present), but no CO in the outer disk either.
- **Debris** (main-sequence stars): gas-poor, cool blackbody-like emission beyond ~10  $\mu\text{m}$  (e.g. Kuiper Belt/zodiacal light), some gas from icy-body evaporation only.

**Q28. What removes disk gas; which process dominates when?**

Two processes: **viscous accretion** onto the central star (removes gas continuously, dominant while the disk is young/dense) and **photo-evaporation** (UV-driven wind stripping outer/surface layers, becomes dominant and drives rapid final clearing once the inner disk has been sufficiently depleted). Total gas disk lifetime  $\sim 10^7$  yr.

**Q29. Derive the vertical density structure; why Gaussian?**

From  $\frac{1}{\rho} \frac{\partial P}{\partial z} = -\Omega^2 z$  (hydrostatic equilibrium, small-angle gravity) and  $P = c_s^2 \rho$  (isothermal ideal gas):  $\frac{1}{\rho} \frac{d\rho}{dz} = -\frac{\Omega^2}{c_s^2} z \Rightarrow \ln \rho = -\frac{\Omega^2}{2c_s^2} z^2 + \text{const} \Rightarrow \rho(z) = \rho_0 \exp\left(-\frac{z^2}{2H^2}\right)$ ,  $H \equiv c_s/\Omega$   
It's Gaussian (not simple exponential like Earth's atmosphere) because the effective vertical gravity *grows linearly* with  $z$  (small-angle projection of the star's radial pull) rather than being constant — pressure support must fight ever-increasing gravity with height, giving a  $z^2$  exponent.

**Q30. Radial force balance equation, term by term.**

$$-\Omega^2 r + \frac{1}{\rho} \frac{\partial P}{\partial r} + \frac{GM}{r^2} = 0$$

- $-\Omega^2 r$ : centrifugal force (outward, from orbital motion).
- $\frac{1}{\rho} \frac{\partial P}{\partial r}$ : pressure-gradient force (outward, since  $P$  decreases with  $r$ ).
- $\frac{GM}{r^2}$ : stellar gravity (inward).

**Q31. Derive the Toomre dispersion relation; meaning of each term.**

Perturb  $\Sigma, P, \Phi, v \rightarrow$  equilibrium  $+\ (\cdot)_1(r, t)$ , with  $\Sigma_1, P_1 \propto e^{i(kr - \omega t)}$  (both spatial and temporal dependence). Linearizing the radial equation of motion + Poisson's equation for self-gravity yields:  $\omega^2 = \Omega^2 - 2\pi G \Sigma_0 k + c_s^2 k^2$

- $\Omega^2$ : **rotation** — stabilizing (shear resists radial contraction).
- $-2\pi G \Sigma_0 k$ : **self-gravity** — destabilizing (drives collapse).
- $c_s^2 k^2$ : **pressure** — stabilizing (resists compression), dominates at small scales.

**Q32. What determines gravitational instability; derive  $k_{\text{crit}}$  and  $Q_{\text{cr}}$ ?**

Instability occurs if  $\omega^2 < 0$  for *some*  $k$  (perturbation grows exponentially,  $\delta P \propto e^{+|\omega|t}$ ). Minimizing  $\omega^2(k)$ :  $d\omega^2/dk = 0 \Rightarrow k_{\text{crit}} = \pi G \Sigma_0 / c_s^2$ . Requiring  $\omega^2(k_{\text{crit}}) \geq 0$  for marginal stability:  $c_s^2 \Omega^2 \geq (\pi G \Sigma)^2$ , i.e.  $Q_{\text{cr}} \equiv \frac{\omega \Omega}{\pi G \Sigma} \geq 1$  (stable)

**Q33. Why does a hot disk resist fragmentation; high  $\Sigma$  promote it?**

$Q_{\text{cr}} \propto c_s \propto \sqrt{T}$ : higher temperature  $\rightarrow$  larger thermal pressure support  $\rightarrow$  larger stabilizing  $c_s^2 k^2$  term dominates over self-gravity, raising  $Q_{\text{cr}}$  above 1 (stable).  $Q_{\text{cr}} \propto 1/\Sigma$ : higher surface density  $\rightarrow$  stronger self-gravity term  $-2\pi G \Sigma k \rightarrow$  lowers  $Q_{\text{cr}}$ , favoring instability. Instability favored by **cold + dense** disks.

**Q34. Toomre criterion applied to the MMSN at Jupiter's orbit.**

At Jupiter's distance, MMSN gives  $\Sigma \approx 10^3 \text{ g cm}^{-2}$ , giving  $Q_{\text{cr}} \approx 3 \times 10^3 / \Sigma \gg 1$  — the Solar System's disk was **stable** against gravitational instability. Conclusion: Jupiter could **not** have formed by direct gravitational collapse ("core collapse") of the gas disk; it must have formed via **core accretion** instead.

**Q35. Role of streaming instabilities vs. Toomre instability?**

Toomre instability is a **global**, disk-scale gravitational collapse mechanism. Streaming instabilities are a **local** aerodynamic mechanism: particle-gas drag feedback concentrates solids into dense clumps even in a globally Toomre-*stable* disk, which can then collapse gravitationally on small scales into planetesimals — but this requires specific conditions on particle size and local metallicity that are not always met.

**Q36. Why must the disk be Toomre-stable for core accretion to be relevant?**

If  $Q_{cr} < 1$ , the disk itself fragments gravitationally into bound clumps directly (core collapse / gravitational instability), forming giant planets (or even stellar companions) essentially instantaneously on dynamical timescales — bypassing the slow, staged solid-body buildup (dust → planetesimals → core → envelope) that defines core accretion. A Toomre-stable disk rules out this fast channel, making the gradual core-accretion route the operative mechanism (consistent with the Solar System's  $Q_{cr} \gg 1$ ).

CHAPTER 1 — Physics of Molecular Clouds

§1.1–1.2

Q37. What evidence first suggested the ISM is not empty?

Barnard (1899) observed star-poor "holes" in photographic plates of the Milky Way, speculating they were filled with "nebulous matter" rather than truly empty. Hartmann (1904) found a stationary Ca absorption line in a spectroscopic binary's spectrum that did *not* share the binary's periodic orbital Doppler shifts — implying it originated in static intervening gas, not the stars themselves.

Q38. Derive the Jeans mass/length; assumptions and their breakdown.

Balance outward pressure acceleration  $\nabla P / \rho_0 \sim P / (\rho_0 r)$  against inward self-gravity  $GM/r^2$  ( $M = \rho r^3$ ), with ideal gas  $P = c^2 \rho_0$ . Collapse requires  $G\rho_0 > c^2/r^2$ , defining:  $r_J = \frac{c}{\sqrt{G\rho}}$ ,  $M_J = \frac{4}{3}\pi\rho_0 r_J^3 = \frac{4\pi c^3}{3\sqrt{G\rho}}$ . Assumes a **uniform, static** medium and subsonic, ideal-gas behavior. Breaks down because real clouds are highly **inhomogeneous** and dominated by **supersonic turbulence** — motivating the virial theorem as a more general (energy-based) stability criterion.

Q39. Derive the virial theorem; meaning of the two terms.

$I = \sum_i m_i \mathbf{r}_i \cdot \mathbf{r}_i$ ; equilibrium means  $\dot{I} = 0$ . Differentiating twice and using  $m_i \dot{\mathbf{v}}_i = \mathbf{F}_i = \sum_{j \neq i} \mathbf{F}_{ij}$ :  $\ddot{I} = 2 \sum_i m_i \dot{\mathbf{v}}_i \cdot \mathbf{r}_i + 2 \sum_i m_i v_i^2 = 0$ . The first term reduces (using  $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$  and Newtonian gravity) to  $2\Omega_G$  (twice the gravitational potential energy); the second is  $4K$  (twice the kinetic energy). Result:  $\boxed{\Omega_G + 2K = 0}$ .

Q40. What is the virial mass; behavior above/below it?

The virial mass is the mass needed for a structure to be self-gravitationally bound in equilibrium:  $M \approx G^{-1} \sigma^2 R$  (uniform sphere case). If actual mass  $> M_{vir}$ , gravity dominates → **collapse**. If mass  $< M_{vir}$ , internal motions dominate → the cloud **disperses** (unless confined by external pressure).

Q41. Larson's scaling relations; link to virial condition and turbulence.

Observed:  $M \sim R^2$  (constant surface density across cloud types) and  $\sigma \sim \sqrt{R}$ . The second follows directly from the virial condition  $M \approx G^{-1} \sigma^2 R$  combined with  $M \sim R^2$ :  $\sigma^2 \sim GM/R \sim GR \Rightarrow \sigma \sim \sqrt{R}$ . Alternatively, this scaling emerges from **compressible turbulence**: structure functions  $S_p(\tau) = \langle |v(r) - v(r + \tau)|^p \rangle \sim \tau^{s_p}$  follow Kolmogorov-type power laws, suggesting turbulence — not simple virial force balance — is the underlying physical driver.

§1.3 Emission-Line Diagnostics

Q42. Derive the high-/low-density limits for a 2-level atom; critical density.

Balance:  $N_1 N_e Q_{12} = N_2 (A_{21} + N_e Q_{21})$ . Two limits:

- **High density** ( $N_e Q_{21} \gg A_{21}$ , LTE):  $N_2/N_1 = (g_2/g_1) e^{-h\nu/k_B T}$  (Boltzmann),  $\epsilon^\ell \propto N_1$  (linear in density).
- **Low density** ( $N_e Q_{21} \ll A_{21}$ ):  $N_2/N_1 = (g_2/g_1) (N_e Q_{21}/A_{21}) e^{-h\nu/k_B T}$ ,  $\epsilon^\ell \propto N_1 N_e$  (quadratic in density).

**Critical density:**  $N_k = A_{21}/Q_{21}$  — the density where the two regimes cross over.

Q43. Why is the H I 21 cm line observable? Origin of 75%/25%?

Despite an extremely small spontaneous decay rate ( $A_{21} = 3 \times 10^{-15} \text{ s}^{-1}$ ,  $\sim 10^7$  yr per atom), the cold ISM's collision rate with electrons, while also low, still exceeds  $A_{21}$  ( $N Q_{12} \sim 10^{-9} \text{ s}^{-1}$ ) — so the gas sits in the **high-density (LTE) regime**. The upper hyperfine level ( $f = 1$ ) has statistical weight  $g = 3$  vs.  $g = 1$  for  $f = 0$ ; with the tiny energy splitting ( $h\nu \ll k_B T$ ), Boltzmann weighting gives population ratio 3:1, i.e. 75% in  $f = 1$ . The line is astronomically weak *per atom* but hugely abundant hydrogen + large column densities make it easily detectable.

Q44. 3-level atom: temperature vs. density diagnostic; example.

$\frac{N_3}{N_2} = \frac{g_3}{g_2} \frac{A_{31}/(N_e Q_{31}) + 1}{A_{31}/(N_e Q_{31}) + 1} \exp\left(-\frac{h(\nu_{31} - \nu_{21})}{k_B T}\right)$ . If  $\nu_{31} \approx \nu_{21}$ : exponential  $\approx 1$ , ratio is **density**-sensitive only. If  $\nu_{31} - \nu_{21}$  is large: ratio dominated by the Boltzmann factor, i.e. **temperature**-sensitive. Example: [O III]  $R = [I(5007) + I(4959)]/I(4363)$  — the auroral  $\lambda 4363$  line (from the much higher  $^1S_0$  level) makes this ratio a robust electron-temperature diagnostic.

Q45. Why do allowed/semi-forbidden/forbidden transitions differ so much in rate?

Allowed (electric dipole) transitions require  $\Delta l = \pm 1$ , parity change — largest transition moment,  $A \sim 10^8 \text{ s}^{-1}$ . Semi-forbidden (dipole with spin flip) violate the spin selection rule weakly. Forbidden (magnetic dipole,  $\Delta l = 0$ , no parity change; or electric quadrupole,  $\Delta l = 0, \pm 2$ ) have vanishing electric-dipole matrix elements and rely on much weaker higher-multipole coupling, since atomic size  $\ll$  wavelength suppresses these terms — rates range down to  $A \sim 10^{-5} - 10^{-15} \text{ s}^{-1}$  for fine/hyperfine forbidden lines.

Q46. LTE vs. non-LTE?

**LTE:** level populations follow the Boltzmann distribution at the local kinetic temperature,  $n_2/n_1 \sim (g_2/g_1) e^{-\Delta E/k_B T}$ ; occurs when collisional de-excitation dominates spontaneous decay ( $n \gg A_{21}/C_{21}$ ). **Non-LTE:** radiative processes compete with/dominate over collisions, populations deviate from Boltzmann at  $T_{kin}$ ; excitation temperature  $T_{ex} < T$  ("sub-thermal"), and line intensity becomes sensitive to density (not just temperature).

§1.3 Chemistry & Evolution

Q47. Why can't H<sub>2</sub> form via two-body gas collisions; role of dust?

Two colliding H atoms have no mechanism to shed the excess binding energy (a third body is needed to carry it away, by momentum/energy conservation) — direct radiative association is far too slow. Dust grain surfaces act as the **third body**: H atoms stick, migrate, and combine on the grain surface, with the grain lattice absorbing the released energy, then  $H_2$  desorbs — this is the dominant  $H_2$  formation channel in the ISM.

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**Q48. Why is CO/ $H_2$  uncertain; why does CO fail as an outer-cloud/disk tracer?**

The ratio depends sensitively on local UV shielding, density, and chemistry, and is observed to range from a canonical  $\sim 10^{-4}$  down to  $\sim 10^{-6}$  in diffuse translucent gas — it is not a universal constant. CO additionally **freezes out** onto grains below  $\sim 17$  K (beyond  $\sim 150$  AU in the Solar System analogy) — in cold outer regions it is simply no longer in the gas phase, so it cannot trace the gas there.

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**Q49. What sets molecular cloud lifetime — H– $H_2$  formation vs. free-fall time?**

H– $H_2$  conversion timescale  $\approx 2.6/n$  [Myr] ( $n$  in units of  $10^3 \text{ cm}^{-3}$ ), comparable to the free-fall time  $t_{ff} \approx 2$  Myr at  $\sim 250 \text{ cm}^{-3}$ . Since **destruction** (UV photodissociation) is faster than **formation**, molecule survival depends on self-shielding: outer cloud layers absorb dissociating photons, protecting the interior — this asymmetry (fast destruction vs. slow, shielding-dependent formation) governs how long a cloud stays chemically “molecular” and star-forming.

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**Q50. Why is the Galactic SFR inconsistent with free Jeans collapse? What provides support?**

If every cloud above  $M_J$  collapsed on its free-fall time, the implied Galactic SFR would far exceed the observed  $\sim 1 M_\odot \text{ kpc}^{-2} \text{ yr}^{-1}$ . This implies self-gravity is **partially counterbalanced** by additional support: primarily **turbulence** and **magnetic fields** (cloud rotation is generally negligible for global support, though relevant later in protostellar contraction).

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**Q51. Hierarchical fragmentation; why does it stop; link to brown-dwarf mass?**

For polytropic collapse ( $P \propto \rho^\gamma$ ),  $M_J \propto \rho^{3(\gamma-4/3)/2}$ . If  $\gamma < 4/3$  (true during the **isothermal** phase,  $\gamma = 1$ ),  $M_J$  *decreases* as density rises, so ever-smaller sub-regions can independently collapse — **hierarchical fragmentation**. It stops (**opacity-limited fragmentation**) once the gas becomes optically thick: radiative cooling becomes inefficient, effective  $\gamma$  rises above  $4/3$ , and thermal pressure halts further fragmentation, setting a minimum fragment mass  $\sim 10^{-2} M_\odot$  — conveniently near the hydrogen-burning limit ( $M_H \approx 0.08 M_\odot$ ), explaining the mass range of brown dwarfs/massive planets as the “floor” of star formation.

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**Q52. Pre-main-sequence evolution: Hayashi line → ZAMS.**

After the main accretion phase, the young star contracts at roughly **constant mass**. Low/intermediate-mass stars first descend nearly **vertically** in the HR diagram (the Hayashi track:  $\sim$ constant, cool effective temperature, decreasing luminosity as radius shrinks) — during this phase the star is fully convective, gravitational contraction supplies the luminosity. As contraction proceeds, the interior heats up; the star then curves and moves toward the **ZAMS**, where core temperature/density finally become sufficient to ignite sustained hydrogen burning, at which point contraction halts and the star settles onto the main sequence. Higher-mass stars traverse this track faster and reach the ZAMS at higher  $T_{eff}$  and  $L$ .

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*End of study guide. 52 questions total across Ch. 1, 2, 4.*